

Cavallaro Leonardo

CONTACT

+39 3932403023

leo.cavallaro.02@gmail.com

leonardo-cavallaro

leo54200

leo54200.github.io

TECHNICAL SKILLS

Languages: Python, Rust, C, Java, SQL, Kotlin, Javascript, ARM Assembly.

Deep Learning & GenAI: PyTorch, TensorFlow, Hugging Face, Stable Diffusion XL (SDXL).

Reinforcement Learning: SAC, REDQ, SynthER, Multi-Agent RL (MARL), SUMO-RL.

Data Science: NumPy, Pandas, Scikit-learn, Matplotlib.

Infrastructure & Tools: Docker, Git/GitHub, React, Tailwind CSS.

LANGUAGES

Italian (Native)

English (Fluent)

PROFILE

Computer Engineering M.Sc. Student at **Polytechnic of Turin**, specializing in **AI & Data Analytics**. Recently completed an exchange at **KAIST**, developing **Generative RL frameworks** that utilize **Diffusion Models** to solve sample inefficiency in complex environments like **SUMO** and **Overcooked-AI**. Proven ability to bridge the gap between high-level **Deep Learning** and **low-level Systems Programming**. Driven by scaling theoretical AI models into high-performance, real-world applications.

WORK EXPERIENCE

10/26 - Present **Research Intern ELLIS Alicante, Spain**
Investigate LLM interpretability by analyzing how negation-based jailbreaks alter layer-wise representations and affect model refusal dynamics.

EDUCATION

09/24 - present **M.Sc. Computer Engineering (AI & Data Analytics)**
Polytechnic of Turin (PoliTO), Italy

09/25 - 02/26 **Exchange Semester**
KAIST, 2nd in S. Korea

10/21 - 07/24 **B.Sc. Computer Engineering**
University of Catania, Italy

02/23 - 07/23 **Erasmus+ Program**
Warsaw University of Technology, 3rd in Poland

TECHNICAL PROJECTS

Generative Reinforcement Learning (KAIST Project)

- Developed a **Diffusion-based generative model (Residual MLP)** to synthesize high-fidelity transitions (s,a,r,s') for **RL replay buffers**.
- Optimized **SUMO (Traffic Control)** agents using a **20:1 Update-to-Data (UTD) ratio**, significantly accelerating convergence in data-limited regimes.
- Investigated **Prioritized Generative Replay (PGR)** using **Intrinsic Curiosity** to guide data generation toward high-uncertainty state regions.
- Conducted a **failure analysis** on discrete multi-agent logic **(Overcooked-AI)**, identifying coordination "hallucinations" in sparse-reward settings as a scaling limitation.

Photomosaic Generation with SDXL (KAIST Project)

- Architected a generative pipeline using **Stable Diffusion XL** and **color quantization** to assemble large images from semantically relevant micro-tiles.
- Implemented **K-Means clustering** for precise tile-matching, balancing visual fidelity with high-dimensional data processing.

AI-Powered Tutor (B.Sc. Thesis)

- Developed a **full-stack orchestration layer** (Laravel and JavaScript) to manage real-time student-AI interactions.
- Designed a **multi-agent tutoring system** leveraging **OpenAI LLM APIs** to create interactive virtual study avatars.

Bare-Metal Pac-Man for ARM Cortex-M3 (PoliTo Project)

- Programmed a hardware-integrated game in **C and ARM Assembly**, implementing manual memory mapping and peripheral interfacing to optimize performance for resource-constrained environments.